

Geometric time-dependent density functional theory*

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General formulation of TDDFT

- ▶ Consider N -particle Hamiltonian $\hat{H}(t) = \hat{T} + \hat{V}_{\text{ee}} + \hat{V}_{\text{ext}}(t)$ and time-dependent Schrödinger equation (TDSE)

$$\begin{cases} i\partial_t \Psi^S(t) = \hat{H}(t) \Psi^S(t) \\ \Psi^S(0) = \Psi_0 \end{cases} \quad \text{leading to the Schrödinger density} \quad \rho^S(t) = \rho_{\Psi^S(t)}.$$

- ▶ Goal of time-dependent density-functional theory (TDDFT) is to find equation involving ρ^S only.

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- ▶ For given $\Psi_0, V_{\text{ext}}(t), \rho(t)$ find operator $\hat{F}(t) = \hat{F}[\Psi_0, V_{\text{ext}}, \rho](t)$ such that

$$\begin{cases} i\partial_t \Psi(t) = \left(\hat{H}(t) + \hat{F}(t) \right) \Psi(t) \\ \Psi(0) = \Psi_0 \end{cases} \quad \text{has the prescribed density} \quad \rho_{\Psi} = \rho.$$

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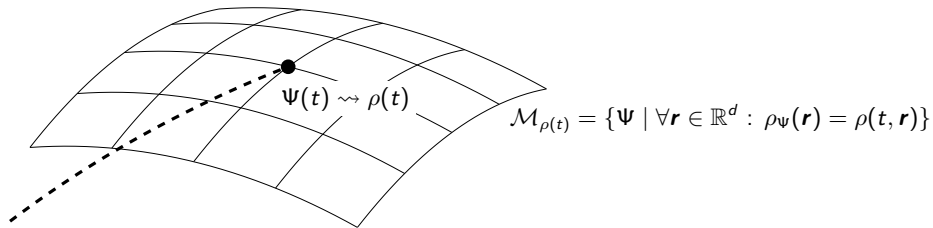
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- ▶ If $\hat{F}$ is **uniquely defined** (within some general class), then $\boxed{\hat{F}[\Psi_0, V_{\text{ext}}, \rho] = 0 \iff \rho = \rho^{\text{S}}}$.
- ▶ Standard TDDFT is the choice $\hat{F}(t) = \sum_{j=1}^N V(t, \mathbf{r}_j)$ a 1-electron potential. Uniqueness of $\hat{F}$ within this class is the **Runge–Gross** (1984) theorem.

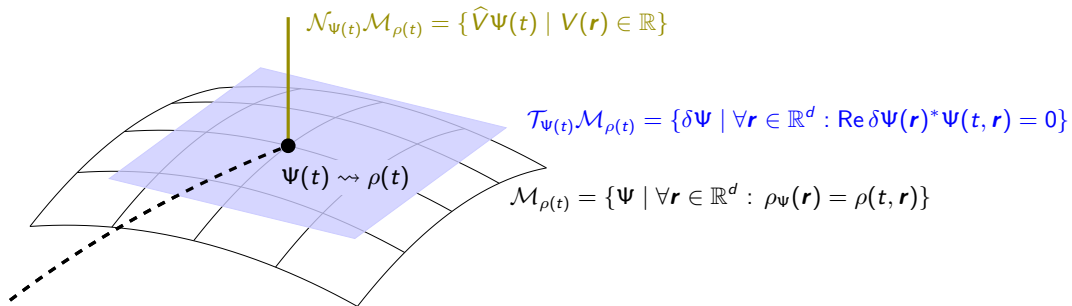
Geometric interpretation of TDDFT

- ▶ Choice of $\hat{F}(t)$ correspond to choice of wave function velocity $\partial_t \Psi(t)$
- ▶ A lot of freedom in this choice if only constraint is to reproduce ρ



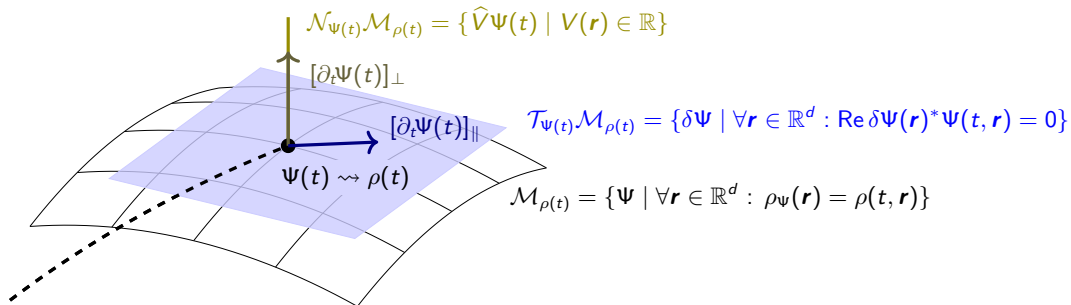
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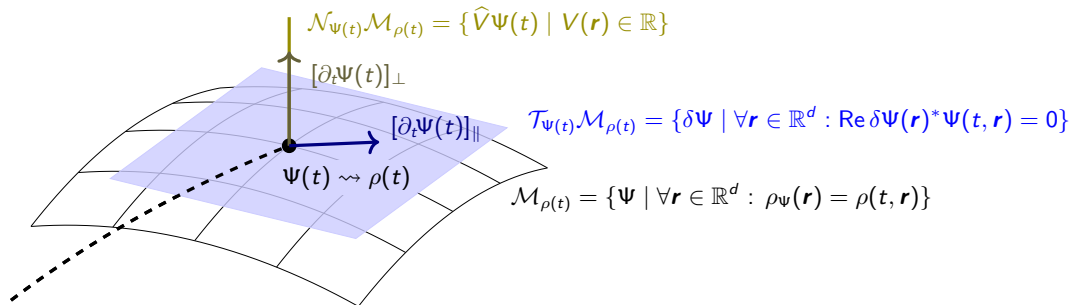


- ▶ Decompose $\partial_t \Psi(t) = [\partial_t \Psi(t)]_{\parallel} + [\partial_t \Psi(t)]_{\perp}$ into tangent and normal component.

Normal component $[\partial_t \Psi(t)]_{\perp}$ is fixed by $\partial_t \rho(t)$. The tangent component $[\partial_t \Psi(t)]_{\parallel}$ is **completely free**

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- ▶ Need additional constraint for how to choose $[\partial_t \Psi(t)]_{\parallel}$. In standard TDDFT this is achieved with the **variational principle**

Standard TDDFT

- ▶ The operator $\hat{F}$ is chosen by the **variational principle**

$$\boxed{\text{stat}_{\Psi \sim \rho} \int_0^T \langle \Psi(t) | (i\partial_t - \hat{H}(t)) \Psi(t) \rangle} \iff \text{Re} \langle \delta\Psi | \underbrace{(i\partial_t - \hat{H}(t)) \Psi(t)}_{\hat{F}(t)\Psi(t)} \rangle = 0 \quad \forall \delta\Psi \in \mathcal{T}_{\Psi(t)} \mathcal{M}_{\rho(t)}$$

- ▶ Thus, $\hat{F}(t)\Psi(t) \in \mathcal{N}_{\Psi(t)} \mathcal{M}_{\rho(t)}$, i.e., $\hat{F}(t) = \sum_{j=1}^N V(t, \mathbf{r}_j)$ with external potential $V(t, \mathbf{r}) \in \mathbb{R}$.

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- ▶ Define **universal functional potential** $V[\Psi_0, \rho] = V[\Psi_0, V_{\text{ext}} = 0, \rho]$. It has $V[\Psi_0, \rho] = V_{\text{ext}} \iff \rho = \rho^S$

- ▶ Consider modified TDSE with this $\hat{F}$

$$\begin{cases} i\partial_t \Psi(t) = (\hat{T} + \hat{V}_{\text{ee}} + \hat{V}[\Psi_0, \rho](t)) \Psi(t) \\ \Psi(0) = \Psi_0 \end{cases} \quad \text{Continuity equation} \quad \partial_t \rho(t) + \nabla \cdot \mathbf{j}_{\Psi(t)} = 0.$$

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- ▶ Differentiating in time we find formula for V in terms of Ψ and ρ (van Leeuwen 1998)

$$\partial_t^2 \rho(t) = -[K_{\rho(t)} V](t, \mathbf{r}) + q_{\Psi(t)}(\mathbf{r}), \quad K_{\rho} V = -\nabla \cdot (\rho \nabla V).$$

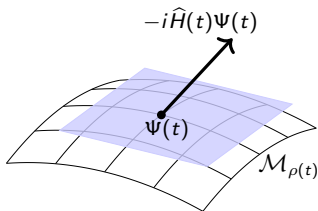
- ▶ TDSE with this V is a highly nonlinear equation and difficult to solve. Mathematically badly behaved. No known results on existence of solutions.

Geometric TDDFT (1/2)

- In geometric TDDFT, $\hat{F}$ is chosen to minimize the norm

$$\boxed{\min_{\partial_t \Psi(t) \leadsto \partial_t \rho(t)} \left\| i \partial_t \Psi(t) - \hat{H}(t) \Psi(t) \right\|} \iff [\partial_t \Psi(t)]_{\parallel} = \hat{P}_{\mathcal{T}_{\Psi(t)} \mathcal{M}_{\rho(t)}} \left(-i \hat{H}(t) \Psi(t) \right)$$

The tangent component $[\partial_t \Psi(t)]_{\parallel}$ is the orthogonal projection of $-i \hat{H}(t) \Psi(t)$ onto the tangent space.

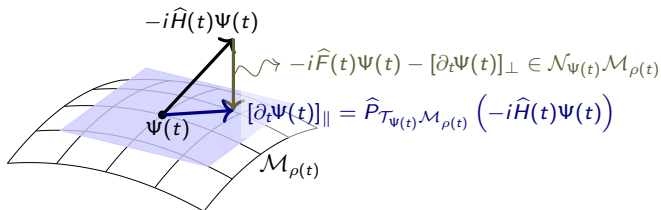


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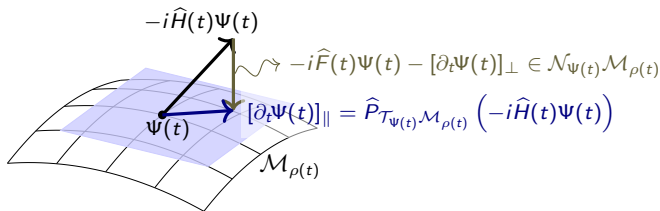
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- Our **geometric Runge–Gross theorem** gives that $W = W[\Psi_0, V_{\text{ext}}, \rho]$ is unique and thus that ρ^S solves

$$W[\Psi_0, V_{\text{ext}}, \rho] = 0 \iff \rho = \rho^S$$

Geometric TDDFT (2/2)

- ▶ Consider modified TDSE with $\hat{F}(t) = \sum_{j=1}^N iW(t, \mathbf{r}_j)$

$$\begin{cases} i\partial_t \Psi(t) = (\hat{H}(t) + i\widehat{W}(t)) \Psi(t) \\ \Psi(0) = \Psi_0 \end{cases} \quad \text{with} \quad \widehat{W}(t) = \sum_{j=1}^N W[\Psi_0, V_{\text{ext}}, \rho](t, \mathbf{r}_j).$$

- ▶ W reproduces ρ and thus preserves the norm: $\langle \Psi(t) | \widehat{W}(t) \Psi(t) \rangle = 0$. Get equivalent **Hermitian form**

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- Continuity equation can be solved to find W in terms of Ψ and ρ . It reads

$$\partial_t \rho(t) + \nabla \cdot \mathbf{j}_{\Psi(t)} = 2S_{\Psi(t)} W(t), \quad [S_{\Psi} W](t, \mathbf{r}) = \rho W + \int \rho_{\Psi}^{(2)}(\mathbf{r}, \mathbf{r}') W(t, \mathbf{r}') d\mathbf{r}'.$$

- W can be viewed as generating local **sources and sinks**

Geometric TD Kohn–Sham (TDKS) (1/2)

- ▶ For $\widehat{V}_{\text{ee}} = 0$ we apply theory to reproduce the density in a **non-interacting** system

$$\begin{cases} i\partial_t \Phi(t) = \left(\widehat{T} + \widehat{V}_{\text{ext}} + i\widehat{W}_s(t) \right) \Phi(t), \\ \Phi(0) = \Phi_0, \end{cases} \quad \rho_{\Phi_0}(\mathbf{r}) = \rho(0, \mathbf{r})$$

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- ▶ Define instead **universal geometric functional** (analogue of Hxc potential)

$$W_{\text{Hxc}}[\Phi_0, \Psi_0, \rho] = W_s[\Phi_0, V[\Psi_0, \rho], \rho], \quad \text{with} \quad V[\Psi_0, \rho] \quad \text{the universal functional potential}$$

We obtain the predictive scheme

$$\begin{cases} i\partial_t \Phi(t) = \left(\widehat{T} + \widehat{V}_{\text{ext}} + i\widehat{W}_{\text{Hxc}}[\Phi_0, \Psi_0, \rho_\Phi](t) \right) \Phi(t), \\ \Phi(0) = \Phi_0, \end{cases}$$

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Proof (This system reproduces ρ^S).

By construction of W_s , we have $W_s[\Phi_0, V_{\text{ext}}, \rho_\Phi] = W_{\text{Hxc}}[\Phi_0, \Psi_0, \rho_\Phi](t) = W_s[\Phi_0, V[\Psi_0, \rho_\Phi], \rho_\Phi]$.
By (geometric) Runge–Gross $V[\Psi_0, \rho_\Phi] = V_{\text{ext}}$ and thus $\rho_\Phi = \rho^S$ by Runge–Gross. □

- Can instead correct any (adiabatic) Hxc approximation $V_{\text{Hxc}}^{\text{app}}[\rho]$:

$$\begin{cases} i\partial_t \Phi(t) = \left(\hat{T} + \hat{V}_{\text{ext}} + \hat{V}_{\text{Hxc}}^{\text{app}}[\rho_\Phi] + i\hat{W}^{\text{corr}}[\Phi_0, \Psi_0, \rho_\Phi](t) \right) \Phi(t), \\ \Phi(0) = \Phi_0, \end{cases}$$

with $W^{\text{corr}}[\Phi_0, \Psi_0, \rho] = W_s[\Phi_0, V[\Psi_0, \rho] + V_{\text{Hxc}}^{\text{app}}[\rho], \rho]$.

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- ▶ Next, study toy systems and compare W to V .
- ▶ Two-electron in singlet state gives general form $i\partial_t \varphi = (-\nabla^2/2 + V + iW)\varphi$. With $\varphi = \sqrt{\rho^S/2} e^{i\theta}$ then

$$W = \frac{\partial_t \rho^S + \nabla \cdot (\rho^S \nabla \theta)}{2\rho^S}, \quad V = -\partial_t \theta - \frac{|\nabla \theta|^2}{2} + \frac{\Delta \sqrt{\rho^S}}{2\sqrt{\rho^S}}.$$

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- ▶ Simple $\theta = 0$ gives the exact adiabatic (eaKS) potential and non-adiabatic (na) W

$$V^{\text{eaKS}} = \frac{\Delta \sqrt{\rho^S}}{2\sqrt{\rho^S}}, \quad W^{\text{na}} = \frac{\partial_t \rho^S}{2\rho^S}.$$

- ▶ In 1D one can find explicitly the non-adiabatic (na) part of V as

$$V^{\text{na}} := V^{\text{eKS}} - V^{\text{eaKS}} = -\frac{1}{2} \left(\frac{j_\varphi}{\rho^S} \right)^2 - \int^x \partial_t \left(\frac{j_\varphi}{\rho^S} \right),$$

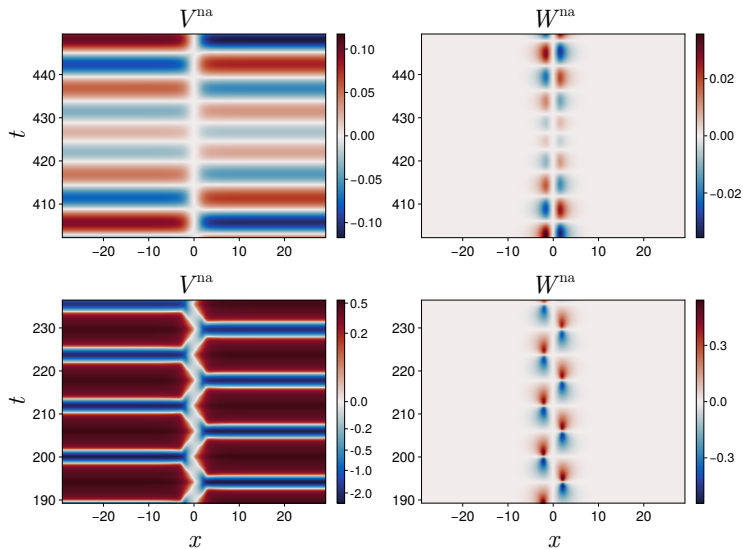


Figure: Rabi oscillations in 1D soft-coulomb $V_{ee}(x) = \frac{1}{\sqrt{1+x^2}}$ in external field $V_{\text{ext}}(t, x) = \frac{-2}{\sqrt{1+x^2}} + xE \sin(\omega t)$

- ▶ (Recap of) geometric perspective of standard TDDFT
- ▶ Novel geometric TDDFT with purely imaginary potential
 - ▶ Based on projecting the Schrödinger dynamics onto manifold of wave functions with prescribed density
 - ▶ Kohn–Sham schemes with only geometric term, or both potential and geometric term
 - ▶ $W^{\text{corr}}[\Phi_0, \Psi_0, \rho] = W_s[\Phi_0, V[\Psi_0, \rho] + V_{\text{Hxc}}^{\text{app}}[\rho], \rho]$ gives exact correction to $V_{\text{Hxc}}^{\text{app}}[\rho]$
- ▶ Solvable toy systems and comparison of potential and geometric term
 - ▶ W^{na} appears more localized than V^{na}
- ▶ Next steps: Construct approximate functionals for $W^{\text{corr}}[\Phi_0, \Psi_0, \rho]$

Thanks for your attention!